Part A · Nine charts, taken apart

All data are synthetic, generated for CDB601220. Figures: left = rebuilt from the recipe, right = honest redesign.

1 · Premium-certified helmets

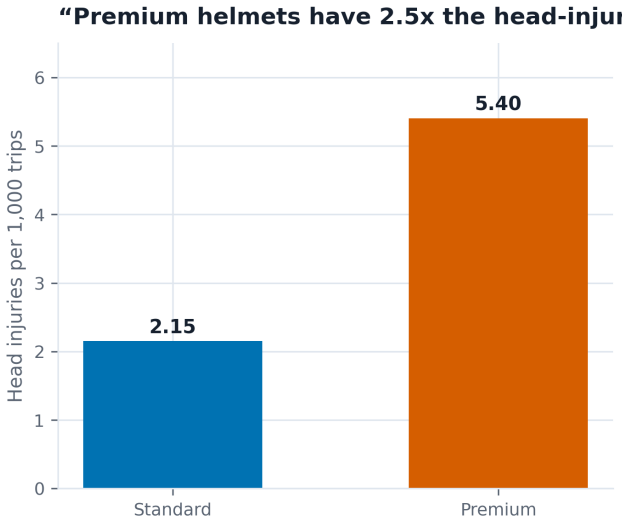
“Premium-certified helmets have 2.5x the head-injury rate” (5.40 vs 2.15 per 1,000 trips)

Conclusion it invites. Premium helmets are worse, perhaps even harmful; riders should stay on Standard.

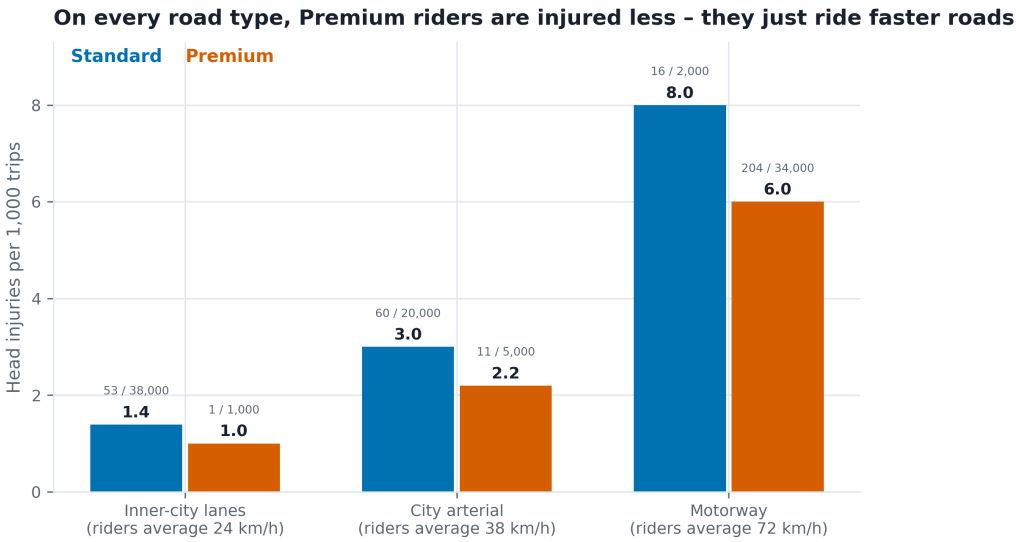
The decoy. Premium riders are simply less experienced. Not so: mean experience is 31.4 months for Premium and 31.5 for Standard.

The mechanism. Simpson’s paradox, with road type as the hidden variable. 340 of the 400 Premium riders (85%) work the Motorway; only 20 of 600 Standard riders (3%) do. Motorway riders average 72 km/h against 24 km/h in the inner city, and the Motorway is where injuries happen (6.1 per 1,000 trips against 1.4 in the lanes). Split by road, Premium is lower every time: 1.0 vs 1.4 (lanes), 2.2 vs 3.0 (arterial), 6.0 vs 8.0 (Motorway). Caveat: Inner-city Premium rests on a single injury, and riders were not randomised, so this says “the 2.5x is not the helmet”, not “Premium works”.

The redesign. Rate per 1,000 trips for each helmet grade within each road type, with injuries/trips printed so small cells are visible.



n = 1,000 riders, 100,000 trips. y-axis from zero. Synthetic data generated for CDB601220.



Grey labels: injuries / trips in each cell (Inner-city Premium rests on 1 injury). n = 1,000 riders. Synthetic data generated for CDB601220.

2 · Chai and quiz failures

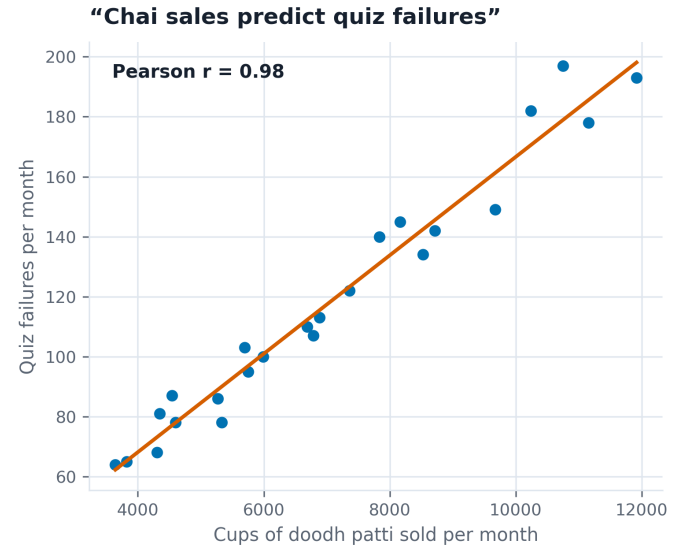
“Canteen chai sales predict quiz failures” (r = 0.98)

Conclusion it invites. Chai drives failures (or at least is a reliable warning sign); the canteen should rethink its menu.

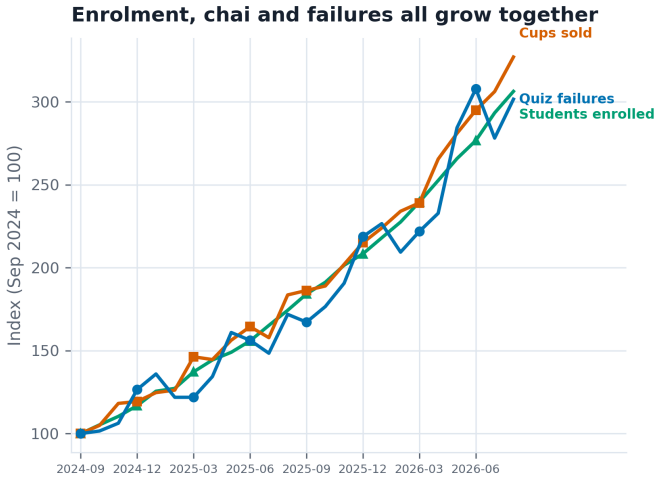
The decoy. Exam months push both up. Not so: r between cups and exam month is only 0.06, and inside regular months r is still 0.996, inside exam months 0.996.

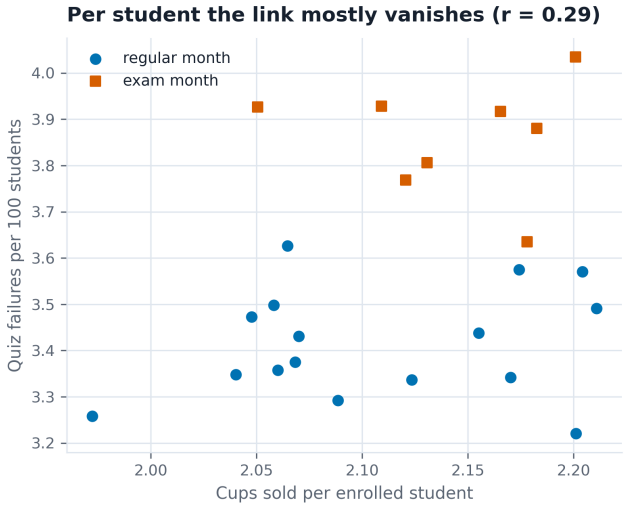
The mechanism. Enrolment growth. Students enrolled grew 3.06x over 24 months (1,765 to 5,406). Enrolment correlates 0.997 with cups sold and 0.977 with failures: more students means more tea and more failed quizzes. Divide both by head-count and r falls from 0.98 to 0.29 (0.20 in regular months, −0.03 in exam months). Exam months do have a higher failure rate per student (3.86 vs 3.42 per 100), but that is a separate effect and unrelated to tea.

The redesign. Left: all three series indexed to Sep 2024 = 100 (a common base, not two axes). Right: per-student rates, circles regular months, squares exam months.



n = 24 months. Synthetic data generated for CDB601220.





n = 24 months. Rates divide by students_enrolled. Synthetic data generated for CDB601220.

3 · The Smog Action Plan

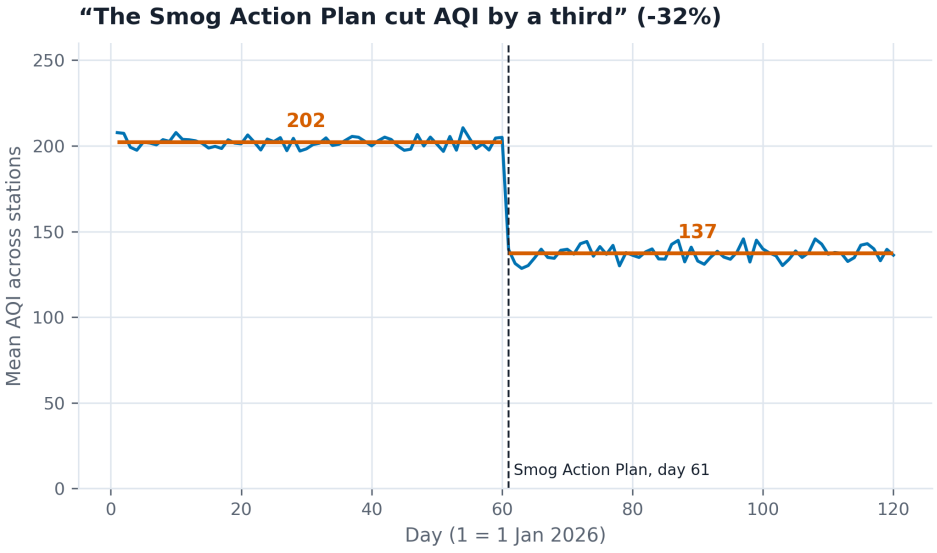
“The Smog Action Plan cut Lahore’s AQI by a third” (202 to 137, −32%)

Conclusion it invites. The plan worked, and by a lot.

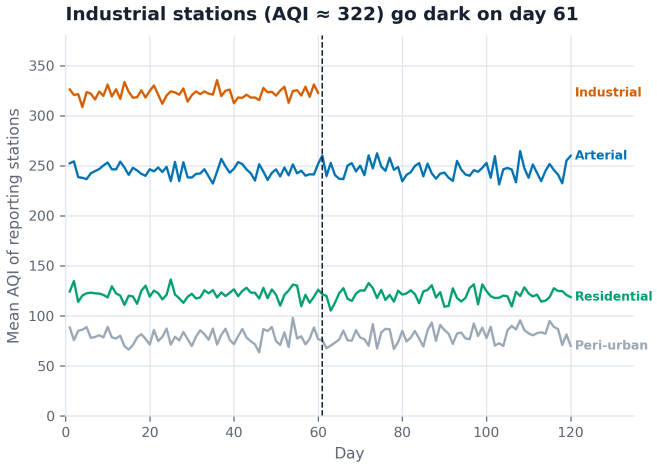
The decoy. The axis exaggerates the fall. Not so: the recipe starts the axis at zero, and the series is flat on both sides with a single cliff at day 61, so it is not a trend or a seasonal slide.

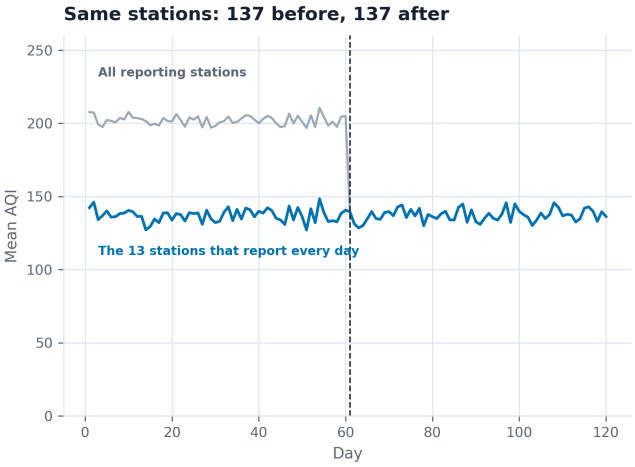
The mechanism. Composition change: the stations that disappeared were the dirtiest ones. All 6 Industrial stations (mean AQI 322) and 3 of the 6 Arterial stations go blank from day 61 (540 blank cells = 9 stations x 60 days). pandas.mean() skips blanks without warning, so the “after” mean is just a cleaner set of stations. On the 13 stations that report all 120 days the mean is 137.1 before and 137.2 after: no change. Every zone mean is also flat (Arterial 244.7 to 245.9, Residential 121.6 to 120.8, Peri-urban 78.9 to 80.4).

The redesign. Left: AQI by zone (Industrial and half of Arterial vanish at the dashed line). Right: all-station mean (grey) against the 13 constant stations (blue).



y-axis from zero. Mean over whichever stations reported. Synthetic data generated for CDB601220.





22 stations x 120 days; 540 station-days blank. y-axes from zero. Synthetic data generated for CDB601220.

4 · Four departments

“Four departments behave identically, so fund them equally”

Conclusion it invites. The four departments have the same study-hours/score relationship (mean 18.0/14.5, r = 0.816, slope 0.25), so treat them alike.

The decoy. n = 11 is too small to tell them apart. It is not the issue: n is equal and identical in all four, and cohort_group maps one-to-one onto department, so there is no hidden intake split either. The sets differ in shape, not noise.

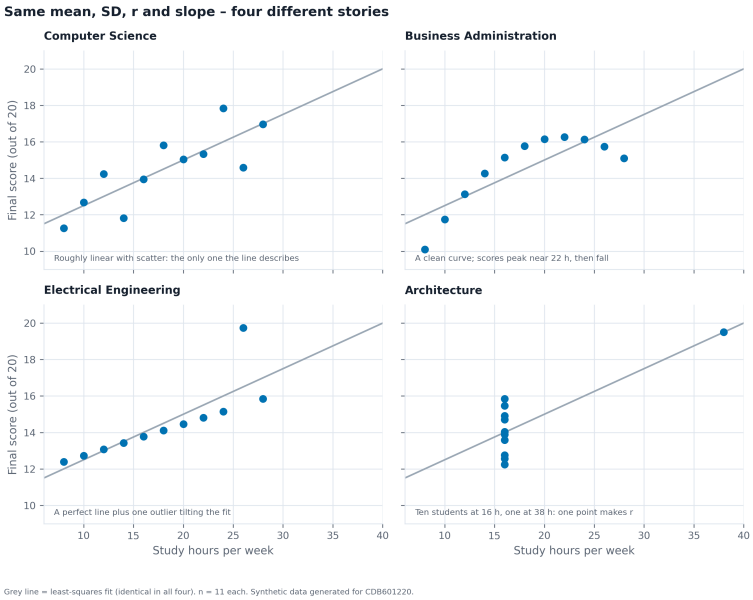
The mechanism. Identical summary statistics, four different shapes (Anscombe). Computer Science is the only ordinary linear scatter. Business Administration is a curve: a quadratic fits at $R^2 = 1.000$ against 0.666 for the line, and scores peak near 22 hours then fall. Electrical Engineering is a perfect line plus one outlier; without it r = 1.000 and the slope is 0.173, not 0.250. Architecture has ten students at exactly 16 hours and one at 38: without that student the SD of hours is 0 and r does not exist, so the whole 0.817 is one point.

The redesign. 2x2 scatter with the identical fitted line in each panel, same axes.

“Four departments behave identically, so fund them equally”

	Computer Science	Business Administration	Electrical Engineering	Architecture
n	11	11	11	11
mean x	18.00	18.00	18.00	18.00
mean y	14.50	14.50	14.50	14.50
SD x	6.32	6.32	6.32	6.32
SD y	1.94	1.94	1.94	1.94
Pearson r	0.816	0.816	0.816	0.817
slope	0.250	0.250	0.250	0.250

11 students per department, n = 44. Synthetic data generated for CDB601220.



Grey line = least-squares fit (identical in all four). n = 11 each. Synthetic data generated for CDB601220.

5 · Five cohorts

“Five cohorts, one set of statistics”

Conclusion it invites. The five cohorts are equivalent: each has r = 0.42 between platform hours and score.

The decoy. Sample size. Not so: every cohort has n = 180, which is plenty, and the statistics match to three decimals because they were forced to (vizlib.match_stats), not by chance.

The mechanism. Summary statistics do not constrain shape. Mean 48.00/63.00, SD 12.00/15.00, r = 0.420 and slope 0.525 are identical, but the cohorts are a noisy line, a ring, a star, two clusters and an X. The sharpest case is the two clusters: inside each cluster the correlation is negative (−0.57 and −0.45); the overall +0.42 comes only from the gap between clusters. In the X, just 22% of points lie within 5 marks of the fitted line.

The redesign. Five scatter plots with the same fitted line; the table in the left figure shows the identical numbers.

“Five cohorts, one set of statistics”

	straight line	ring	star	two clusters	the letter X
n	180	180	180	180	180
mean x	48.00	48.00	48.00	48.00	48.00
mean y	63.00	63.00	63.00	63.00	63.00
SD x	12.00	12.00	12.00	12.00	12.00
SD y	15.00	15.00	15.00	15.00	15.00
Pearson r	0.420	0.420	0.420	0.420	0.420
slope	0.525	0.525	0.525	0.525	0.525

180 students per cohort, n = 900. Synthetic data generated for CDB601220.



n = 180 per cohort, 900 in all. Synthetic data generated for CDB601220.

6 · Creatinine and HbA1c

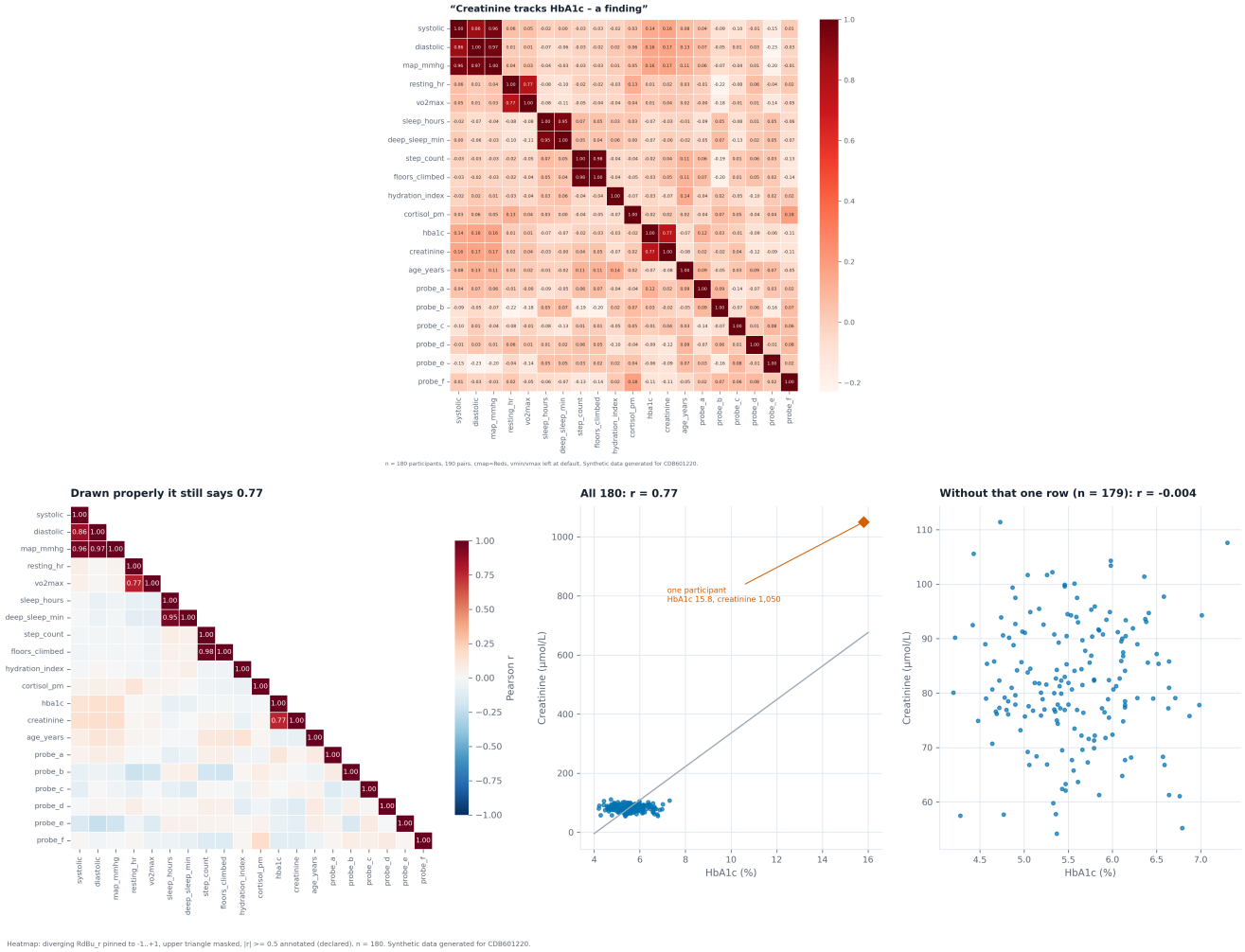
“Creatinine tracks HbA1c – a finding” (r = 0.77)

Conclusion it invites. Kidney function (creatinine) and glycaemic control (HbA1c) move together across the cohort.

The decoy. With 190 pairs, something was bound to look significant. Not here: 20 pairs pass p < 0.05 against about 9.5 expected, but this one has p ≈ 7e-36, nowhere near a chance finding.

The mechanism. One participant. Row 0 has HbA1c 15.8 and creatinine 1,050 (medians are 5.5 and 81.6). That single point is a leverage point: remove it and r = −0.004 (Spearman 0.015); even with it in, the rank correlation is only 0.03. It sits at Site C, the only site with a strong r (0.92). One row of 180 manufactures the headline. (Whether that participant is a real very sick patient or a data error cannot be told from this file.) **Two problems, only one of which the redraw fixes.** The drawing: Reds with no vmin/vmax runs from −0.23 to 1, so negative correlations are pale like zeros and sign is lost, and both triangles show everything twice. A diverging map pinned to −1…+1 with the upper triangle masked fixes the drawing, but it still prints 0.77. Only looking at the scatter exposes the outlier.

The redesign. Left: the corrected heatmap (RdBu_r, −1 to +1, upper triangle masked). Middle: all 180 with the one row marked. Right: the other 179.



7 · ShaheenGo

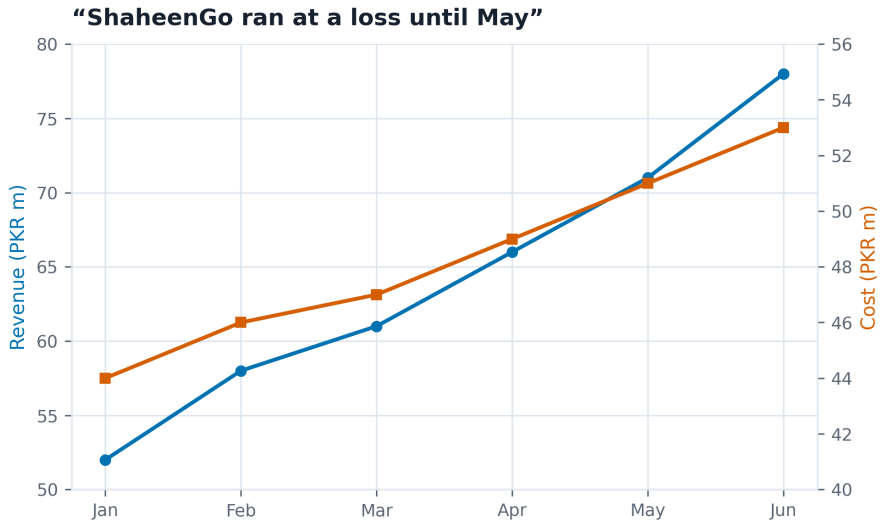
“ShaheenGo ran at a loss until May”

Conclusion it invites. Cost exceeded revenue for the first four months; the company turned profitable in May.

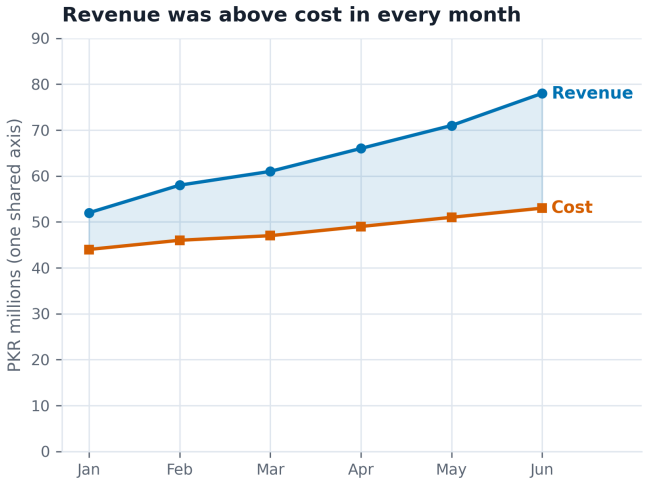
The decoy. That the company really did lose money early. Not so: revenue exceeds cost in every one of the six months (surplus 8, 12, 14, 17, 20, 25 PKR m). Both columns are in PKR millions, so they can be compared directly.

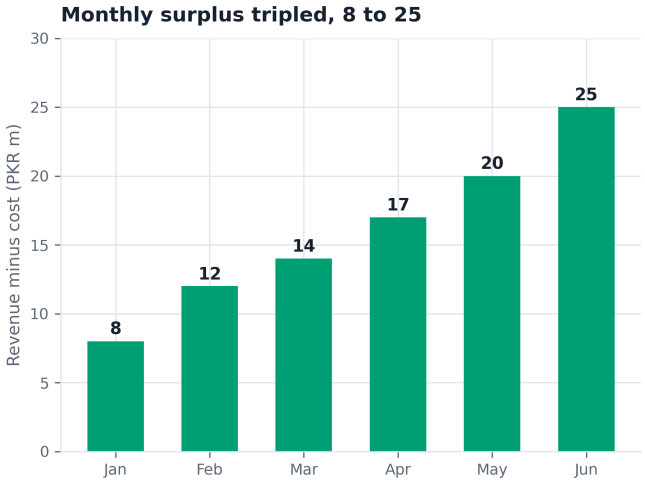
The mechanism. Two y-axes with hand-picked limits. Revenue is drawn on 50–80 and cost on 40–56, so the picture depends on the limits rather than the data. In January the cost point (44) lands at 25% of the plot height and the revenue point (52) at 7%, so cost appears above revenue for Jan–Apr. Over the half-year revenue grew 50% and cost 20%, so the surplus tripled. Both measures share one unit, so there was never a reason for a second axis.

The redesign. Left: both lines on one shared PKR axis from zero. Right: the surplus itself.



Two y-axes with hand-picked limits (50-80 and 40-56). n = 6 months. Synthetic data generated for CDB601220.





Both series are PKR millions, so they share one axis. n = 6 months. Synthetic data generated for CDB601220.

8 · Bin widths

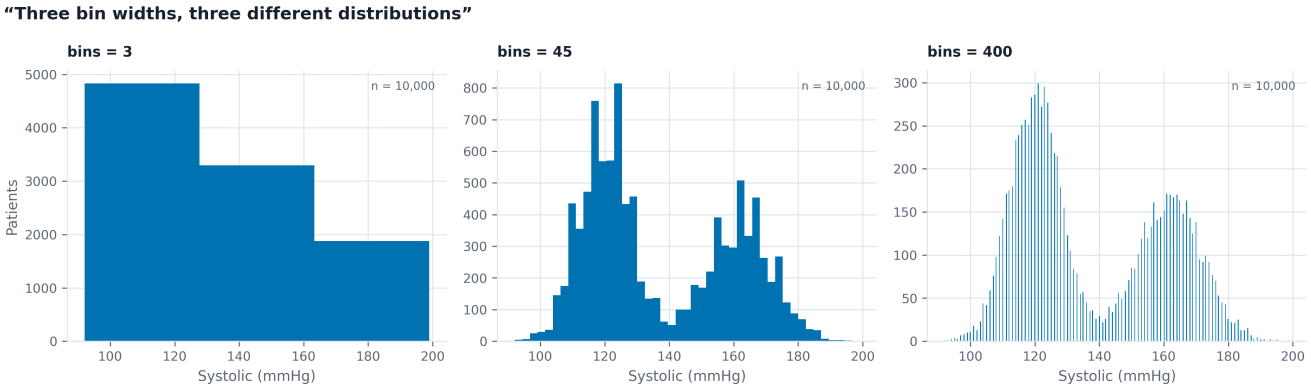
“Three bin widths, three different distributions” (bins = 3, 45, 400)

Conclusion it invites. The shape of systolic pressure depends on who plots it: one block, a few peaks, or a spiky mess.

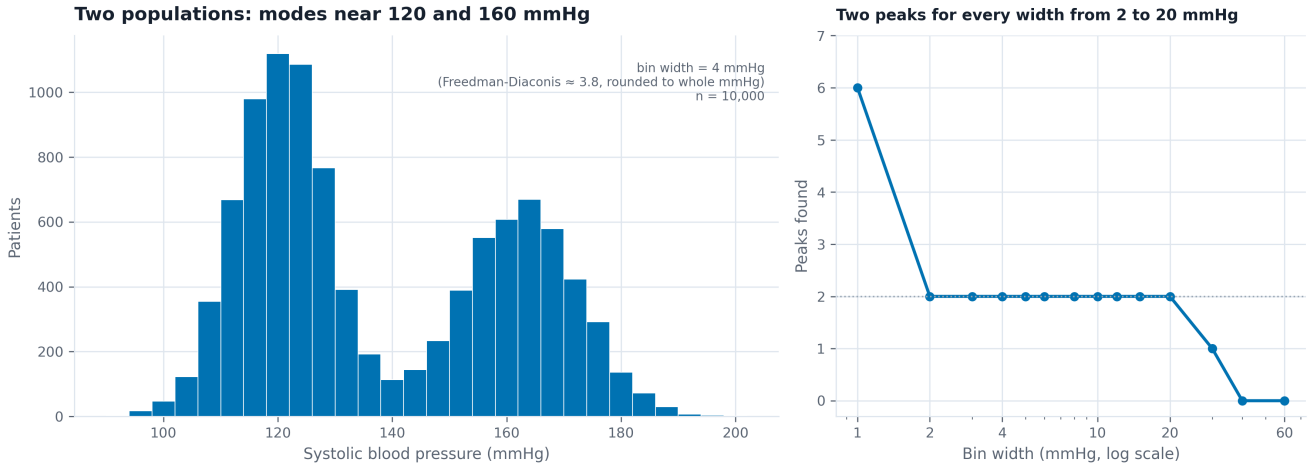
The decoy. Three different samples or too little data. Not so: it is the same 10,000 values in all three panels.

The mechanism. Bin width is the only thing that changed, and two of the three choices are bad. bins = 3 gives bins 35.7 mmHg wide and hides both humps. bins = 400 gives bins 0.27 mmHg wide over whole-number data (104 distinct values), so 296 of the 400 bins are empty and the rest are a comb. bins = 45 gives a width of 2.38, not a whole number, so bars alternate between holding two and three integer values and invent spikes (7 peaks). The truth is steady: with bins aligned to whole mmHg, every width from 2 to 20 shows exactly two peaks (about 121 and 161).

The redesign. Left: width 4 mmHg (Freedman-Diaconis gives 3.8). Right: peaks found against bin width.



Same 10,000 raw systolic values in every panel. Synthetic data generated for CDB601220.



Peaks: scipy.signal.find_peaks, prominence 5% of tallest bin; bins aligned to whole mmHg. n = 10,000. Synthetic data generated for CDB601220.

9 · SpO_2 density

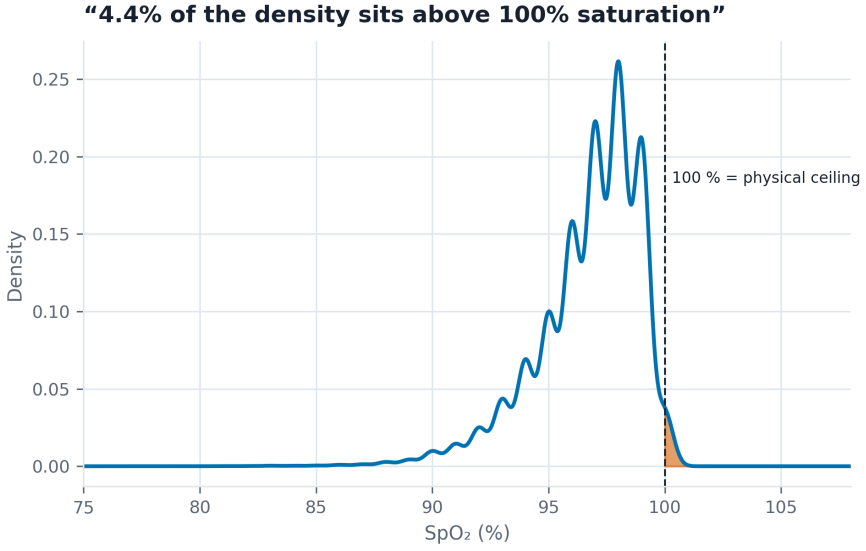
“4.4% of the density sits above 100% saturation”

Conclusion it invites. A material share of patients have impossible oxygen-saturation readings, so the monitors or the records are unreliable.

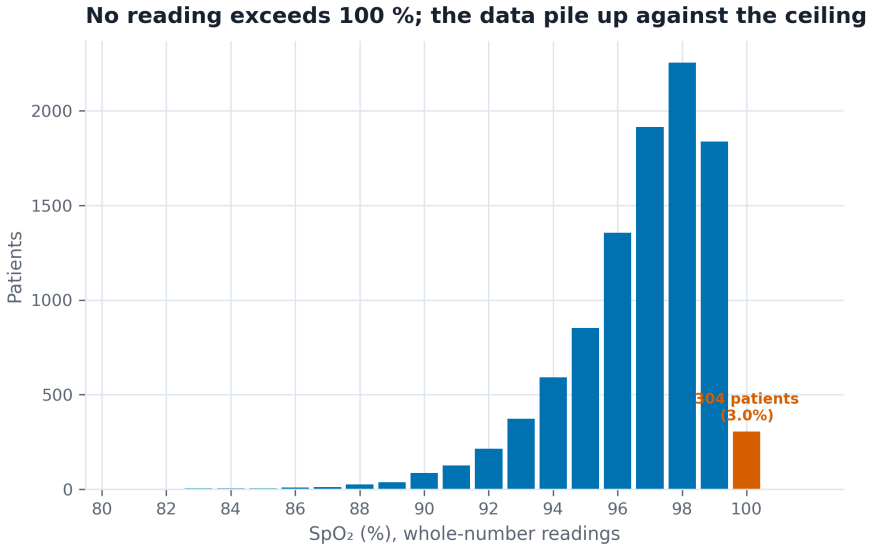
The decoy. Faulty readings above 100. Not so: the maximum in the data is exactly 100 and no row exceeds it. The curve exceeds it; the data do not.

The mechanism. Kernel smoothing across a hard boundary. A KDE puts a Gaussian bump on every point, and a Gaussian has no edge, so mass spills past the physical ceiling. It is worst here because 304 patients (3.0%) sit exactly on 100: the data are piled against the wall. The default (Scott) bandwidth is 0.35 points, narrower than the 1-point spacing of whole-number readings, which also makes the curve lumpy. **Note:** with this recipe I measure 1.6% of the density above 100 (4.3% above 99.5), not 4.4%, so I could not reproduce the headline figure; the mechanism holds either way.

The redesign. One bar per recorded whole-number value, so nothing can be smoothed past 100.



Gaussian KDE, default bandwidth (Scott). Data maximum = 100. n = 10,000. Synthetic data generated for CDB601220.



One bar per recorded integer value, so nothing can be smoothed past 100. n = 10,000. Synthetic data generated for CDB601220.